

DATABASE MANAGEMENT SYSTEMS (DBMS) SYLLABUS

UNIT I

- Database System Applications
- Purpose of Database Systems
- View of Data
- Data Abstraction
- Instances and Schemas
- Data Models
- ER Model
- Relational Model
- Database Languages (DDL, DML)
- Database Architecture
- Storage Manager
- Query Processor
- ER Diagrams
- Entities, Attributes and Relationships
- Relational Model Structure
- Database Schema and Keys

UNIT II

- Relational Query Languages
- Relational Operations
- Relational Algebra
- Selection and Projection
- Set Operations
- Joins
- Division
- Relational Calculus
- SQL Basics
- Aggregate Functions
- GROUP BY
- HAVING
- Nested Queries
- Views
- Triggers

UNIT III

- Normalization
- Functional Dependencies
- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- BCNF
- Multi-valued Dependencies
- Fourth Normal Form (4NF)
- Fifth Normal Form (5NF)

UNIT IV

- Transaction Concepts
- Transaction States
- Atomicity and Durability
- Concurrent Execution
- Serializability
- Recoverability
- Isolation
- Lock-Based Protocols
- Timestamp Protocols
- Validation Protocols
- Recovery Techniques
- Log-Based Recovery
- Buffer Management
- Remote Backup Systems

UNIT V

- File Organization
- Indexing
- Query Processing
- Query Cost Estimation
- Selection Operation
- Projection Operation
- Join Operation
- Aggregate Operation
- Query Optimization
- SQL Query Optimization
- Equivalence Rules